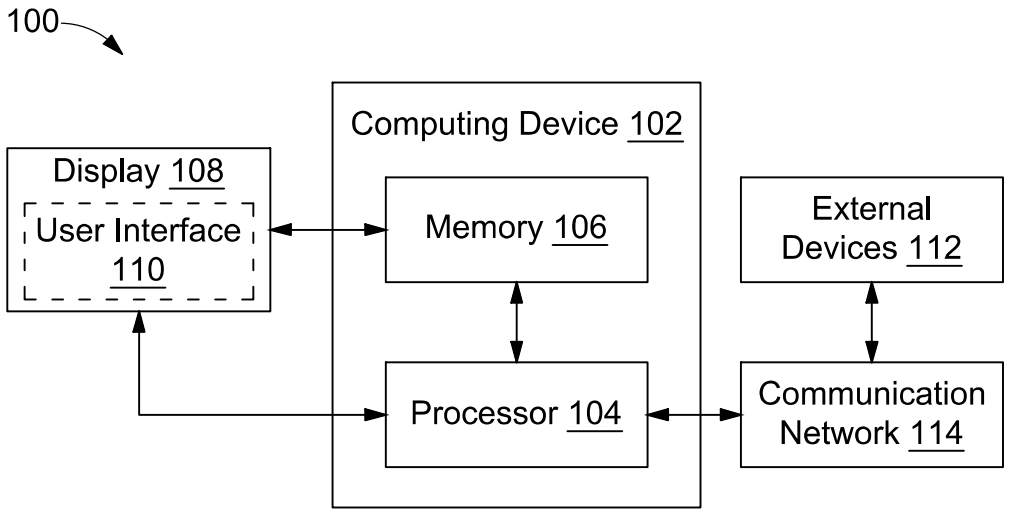

100
Computing Device 102
Display 108
User Interface 110
Memory 106
External Devices 112
Processor 104
Communication Network 114

FIG. 1

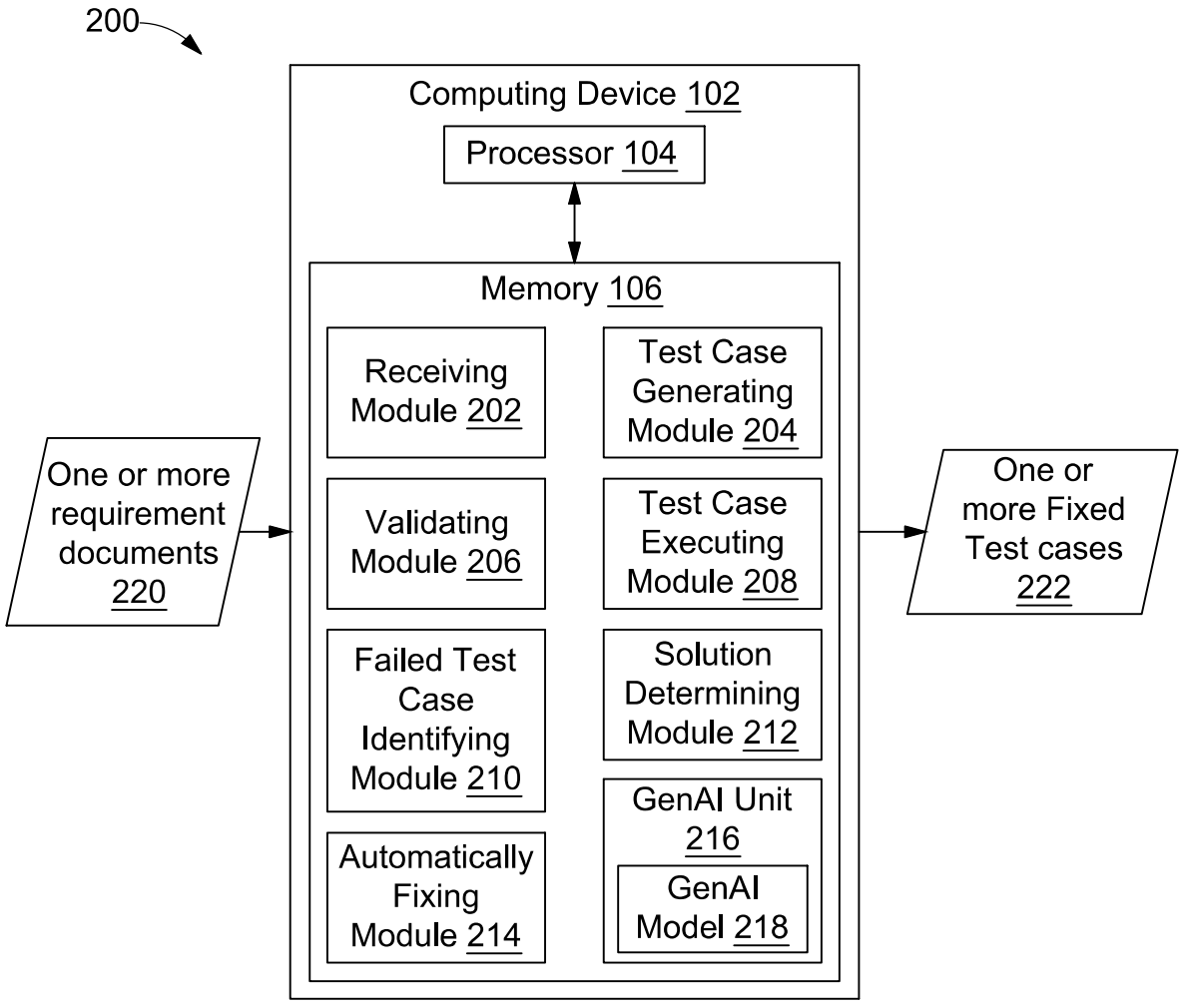

200
Computing Device 102
Processor 104
Memory 106
Receiving Module 202
Test Case Generating Module 204
Validating Module 206
Test Case Executing Module 208
Failed Test Case Identifying Module 210
Solution Determining Module 212
Automatically Fixing Module 214
GenAI Unit 216
GenAI Model 218
One or more requirement documents 220
One or more Fixed Test cases 222

FIG. 2

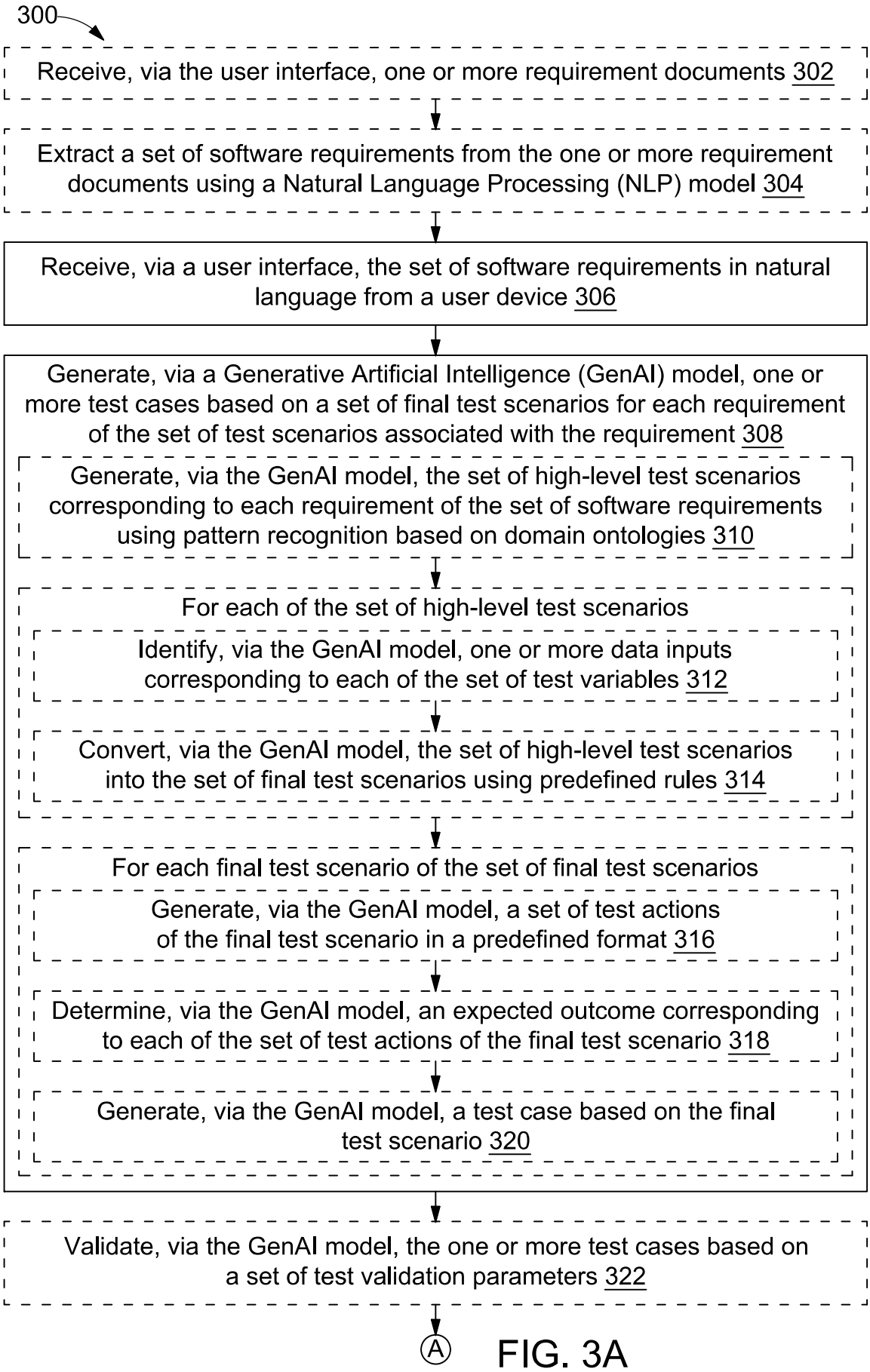

300
Receive, via the user interface, one or more requirement documents 302
Extract a set of software requirements from the one or more requirement documents using a Natural Language Processing (NLP) model 304
Receive, via a user interface, the set of software requirements in natural language from a user device 306
Generate, via a Generative Artificial Intelligence (GenAI) model, one or more test cases based on a set of final test scenarios for each requirement of the set of test scenarios associated with the requirement 308
Generate, via the GenAI model, the set of high-level test scenarios corresponding to each requirement of the set of software requirements using pattern recognition based on domain ontologies 310
For each of the set of high-level test scenarios
Identify, via the GenAI model, one or more data inputs corresponding to each of the set of test variables 312
Convert, via the GenAI model, the set of high-level test scenarios into the set of final test scenarios using predefined rules 314
For each final test scenario of the set of final test scenarios
Generate, via the GenAI model, a set of test actions of the final test scenario in a predefined format 316
Determine, via the GenAI model, an expected outcome corresponding to each of the set of test actions of the final test scenario 318
Generate, via the GenAI model, a test case based on the final test scenario 320
Validate, via the GenAI model, the one or more test cases based on a set of test validation parameters 322
A

FIG. 3A

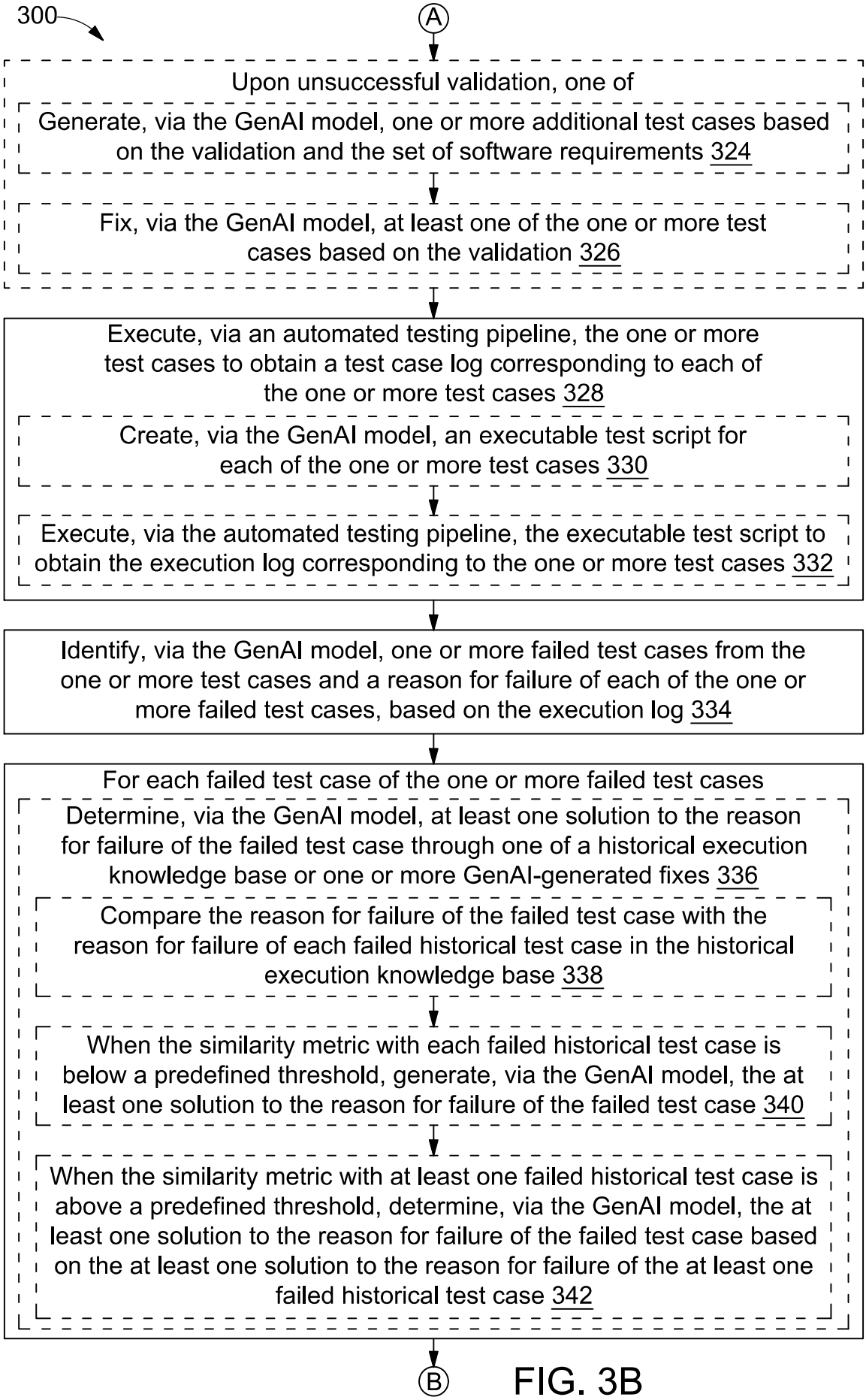

300
A
Upon unsuccessful validation, one of
Generate, via the GenAI model, one or more additional test cases based on the validation and the set of software requirements 324
Fix, via the GenAI model, at least one of the one or more test cases based on the validation 326
Execute, via an automated testing pipeline, the one or more test cases to obtain a test case log corresponding to each of the one or more test cases 328
Create, via the GenAI model, an executable test script for each of the one or more test cases 330
Execute, via the automated testing pipeline, the executable test script to obtain the execution log corresponding to the one or more test cases 332
Identify, via the GenAI model, one or more failed test cases from the one or more test cases and a reason for failure of each of the one or more failed test cases, based on the execution log 334
For each failed test case of the one or more failed test cases
Determine, via the GenAI model, at least one solution to the reason for failure of the failed test case through one of a historical execution knowledge base or one or more GenAI-generated fixes 336
Compare the reason for failure of the failed test case with the reason for failure of each failed historical test case in the historical execution knowledge base 338
When the similarity metric with each failed historical test case is below a predefined threshold, generate, via the GenAI model, the at least one solution to the reason for failure of the failed test case 340
When the similarity metric with at least one failed historical test case is above a predefined threshold, determine, via the GenAI model, the at least one solution to the reason for failure of the failed test case based on the at least one solution to the reason for failure of the at least one failed historical test case 342
B

FIG. 3B

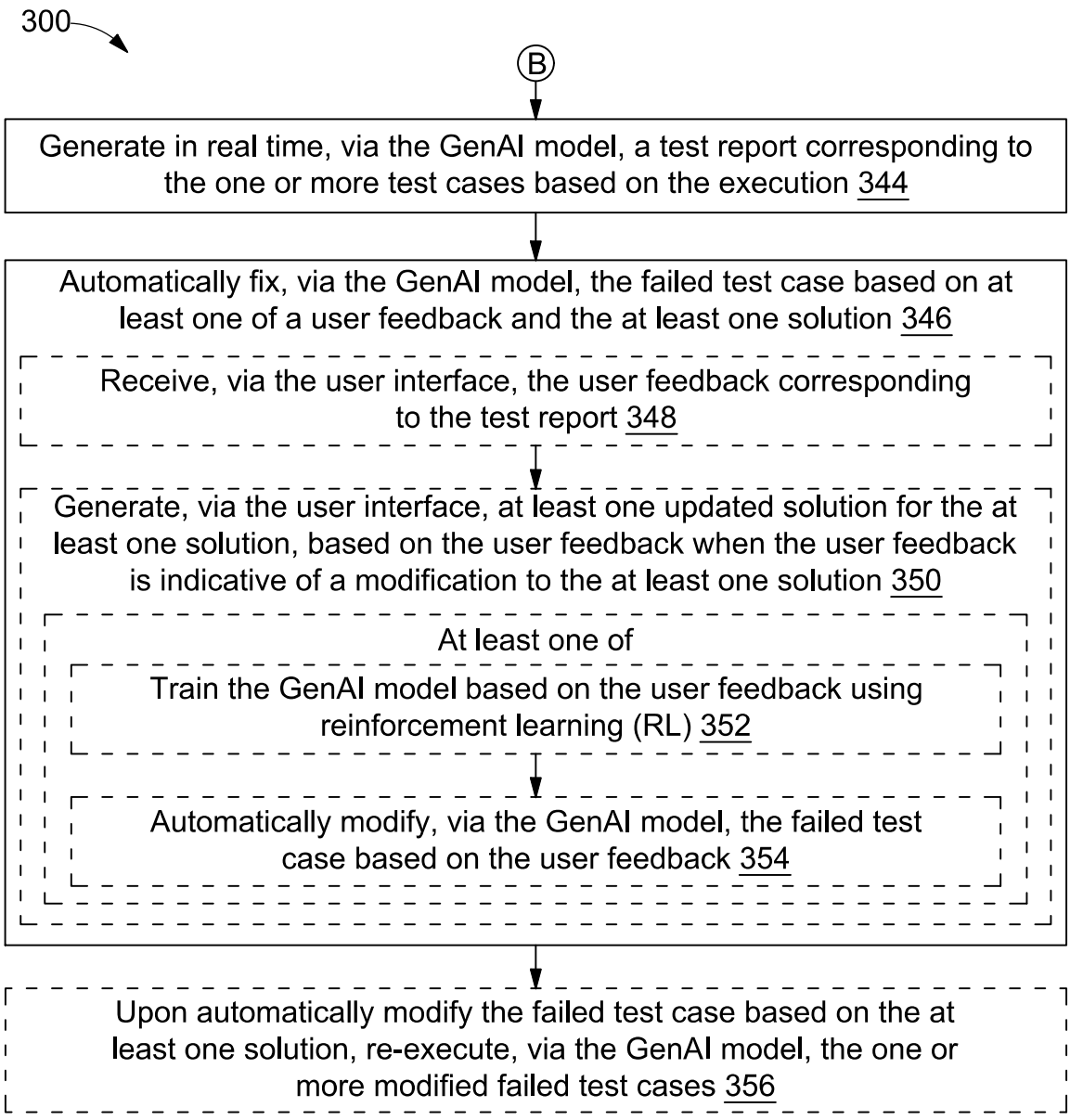

300
B
Generate in real time, via the GenAI model, a test report corresponding to the one or more test cases based on the execution 344
Automatically fix, via the GenAI model, the failed test case based on at least one of a user feedback and the at least one solution 346
Receive, via the user interface, the user feedback corresponding to the test report 348
Generate, via the user interface, at least one updated solution for the at least one solution, based on the user feedback when the user feedback is indicative of a modification to the at least one solution 350
At least one of
Train the GenAI model based on the user feedback using reinforcement learning (RL) 352
Automatically modify, via the GenAI model, the failed test case based on the user feedback 354
Upon automatically modify the failed test case based on the at least one solution, re-execute, via the GenAI model, the one or more modified failed test cases 356

FIG. 3C

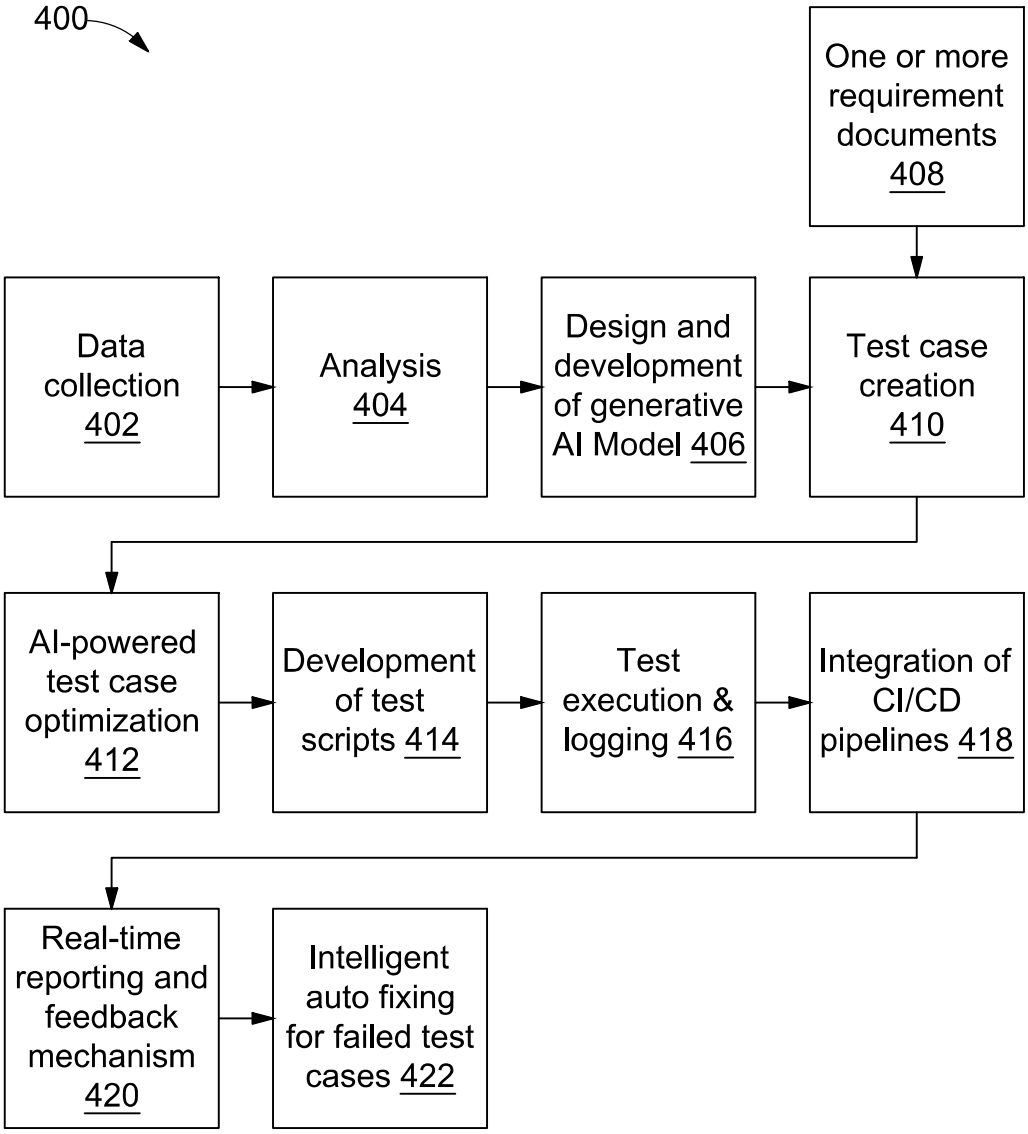

400
One or more requirement documents 408
Data collection 402
Analysis 404
Design and development of generative AI Model 406
Test case creation 410
AI-powered test case optimization 412
Development of test scripts 414
Test execution & logging 416
Integration of CI/CD pipelines 418
Real-time reporting and feedback mechanism 420
Intelligent auto fixing for failed test cases 422

FIG. 4

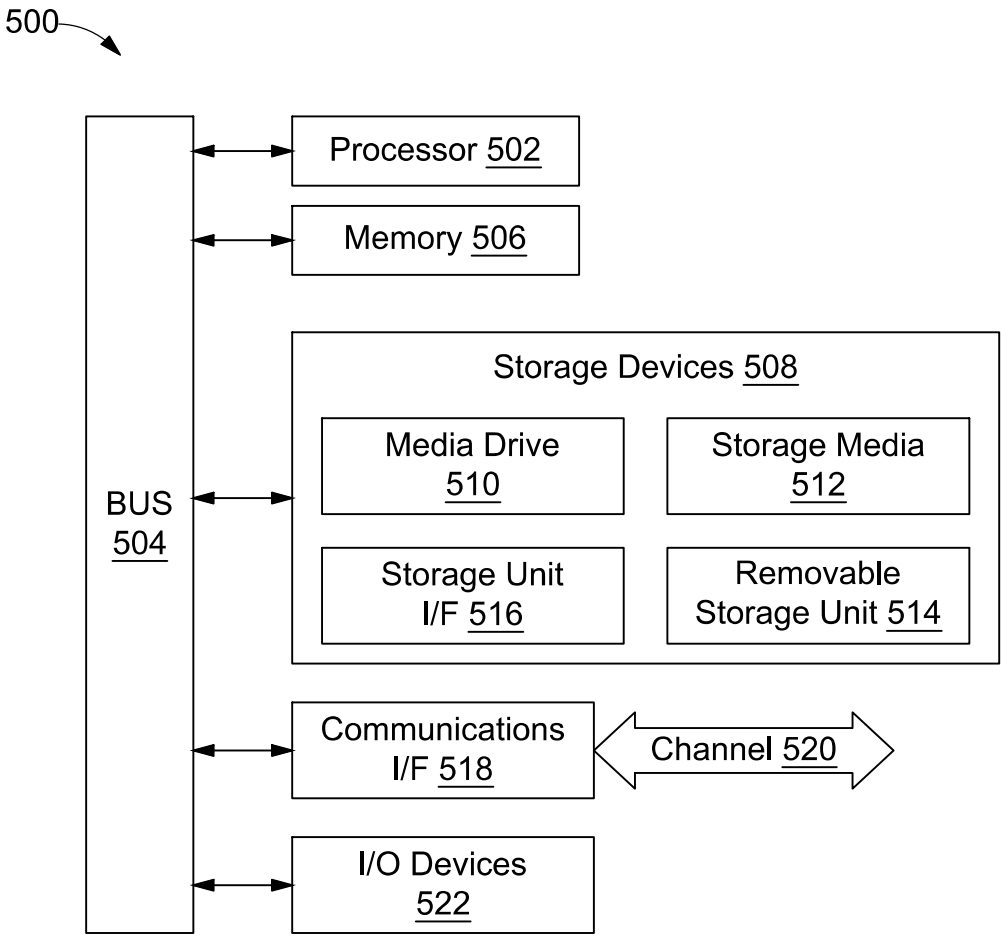

500
Processor 502
Memory 506
Storage Devices 508
Media Drive 510
Storage Media 512
Storage Unit I/F 516
Removable Storage Unit 514
BUS 504
Communications I/F 518
Channel 520
I/O Devices 522

FIG. 5